

Time Chamber: Creating a Virtual Environment for Time Dilation with the Selection of Liking Temporal Cues

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Abstract

Time perception is proven to be influenced by various factors, yet integrating these factors into a cohesive experience remains challenging. Our research explores creating a virtual environment where participants engage in short-term activities while experiencing a lengthened subjective sense of time. In the pilot study, we identified items expected to slow down subjective time perception. Next, in the user study, we create virtual environments consisting of their decision of 'liking' and 'selection' to determine their mental measurements and time perception. The result showed that participants 'liking' and 'selection' of items in the VE influenced their pleasure and focus. However, these factors did not significantly alter their perception of time. This finding underscores the intricacy involved in designing time-dilating VEs. Our next step will focus on designing a more nuanced approach to filtering and integrating environmental elements to manipulate time perception preciously.

CCS Concepts

• **Human-centered computing** → *Interaction design process and methods.*

Keywords

Virtual Reality, Time Perception, Time Dilation, Temporal Cue

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1 Introduction

"Time flies when you're having fun" [1] describes how time seems to pass more rapidly during enjoyable experiences. Time perception encompasses subjective time, one's feeling of time passing, and objective time, which relates to real-time. Extensive research has identified various factors, including emotions [26, 33, 37], colors [9, 31, 32], and music [5, 23], that contribute to the dilation of subjective time, making it appear longer than objective time. Additionally, studies [3, 16, 21] highlight the influence of attention on time perception, indicating that more attention to complex stimuli can make subjective time feel prolonged. Controlling user attention is thus one way to achieve a slower subjective time in virtual reality.

However, transforming these factors into effective interaction techniques and application designs remains challenging. Each factor has trade-offs; for example, negative emotions often lead to time dilation [33], but designing interactions with negative emotions is undesirable. Time cue items can induce time dilation but might heighten the awareness of time passing in virtual space. Moreover, environments combining liking and time dilation are highly subjective, necessitating customization. Selection has also been mentioned to enhance users' engagement, and influenced their time sensation [5]. Not all interaction techniques can consistently maintain positive emotions. Therefore, we aim to explore if it is possible to create a "happiness time" that is subjectively lengthened using these factors.

Our research explores the design considerations and limitations in creating a virtual environment (VE) that fosters prospective time dilation. We aim to enable participants to remain focused and maintain positive emotions in short-term activities. In our pilot study, we designed virtual items hypothesized to extend subjective time perception. Our user study then examined if these items, combined with liking and selection elements, could influence emotions and focus in a VE with time-dilated properties. Contrary to expectations, time perception remained unaffected. We suggest that the items used lacked sufficient intensity to significantly alter time perception. Future research should focus on more robust designs to achieve pronounced time dilation in VE.

2 Related Work

2.1 Time Dilation Research

Time illusion can be divided into time compression and time dilation. Time compression refers to the subjective feeling of time passing more quickly than objective time, while time dilation refers to the subjective feeling of time passing more slowly. According to Tipples's study, negative emotions are more likely to cause time dilation [33]. This phenomenon is common in situations where the user is waiting, as their focus on waiting amplifies the passage of time. For example, when staring at a moving second hand, the user's attention on the clock can make time feel like it is moving slowly, resulting in subjective time dilation.

Factors that can achieve a sense of time dilation include: (a) Music: Rogers et al. [23] found that gameplay experiences without music resulted in smaller emotional fluctuations and increased subjective time perception. Cassidy and Macdonald [5] observed that low-arousal music reduced positive emotions, causing time dilation, while Lake et al. [13] noted that lower pitch likely slowed down time. (b) Environmental Color: Research shows that blue stimuli are perceived to last longer than red stimuli [32]. Shibasaki and Masataka [31] found that males feel subjective time is longer in red light compared to females. Aschoff and Daan [2] discussed how different brightness levels influence time perception. (c) Scene Changes: Glicksohn et al. [8] found that simpler environments increased users' attention and subjective time. (d) Object Movement Speed and Weight: Bruder et al. [4] showed that wearing heavy equipment and experiencing slowed screen movement in virtual reality gave users the illusion of slowed subjective time. Wollner et al. [38] demonstrated that slow motion can lead to time dilation compared to real speed. (e) Attention: Brown's study [3] indicated that the more attention participants paid to an audio file, the longer their subjective time felt. Ornstein's theory [21] suggests that focusing on more complex objects extends subjective time perception. Additionally, different focus types, such as interoceptive, retrospective [22], prospective, and retrospective [3, 34], impact time perception differently.

2.2 Time perception in Virtual Reality

When experiencing virtual reality (VR), the immersive head-mounted display will block out external information, including time information, leading to a subjective time illusion. Hence, some research concentrates on time perception in VR. Similar to research measuring time perception in the real world, studies have explored the relationship between emotion and time perception in VR [12, 37]. In addition, some have investigated how the VR representation of one's body can alter time perception [35, 36]. Conversely, other research has examined different scenarios in VR experiences. For instance, Schneider et al. [28, 29] discussed different patients receiving chemotherapy and found that VR during chemotherapy leads to a perceived compression of elapsed time, showing VR's potential to make chemotherapy treatments more tolerable. Rutrecht et al. [25] compared the flow state in rhythm games played in VR versus on a computer screen, confirming that flow states in VR lead to a subjective speeding up of time and better game performance. Similarly, Mullen et al. [19] also find that VR leads to time compression in waiting time compared to the computer screen. Some researchers

utilize virtual objects to influence subjects' time perception. Research pointed out that larger visual changes on virtual features might lead to time compression [17]. The impact of virtual zeitgebers on time estimation and how different types of moving objects influence time perception have been discussed [15, 27]. Landeck et al. [14] designed high-speed and high-density virtual tunnels to explore the subject's time perception, showing that subjective time passed faster under virtual tunnels. This suggests that virtual tunnel simulation can manipulate time perception in VR.

However, most research in VR time perception has focused on time compression. Our study, in contrast, aims to create a VE that induces a sense of time dilation.

3 Research Questions

To evoke pleasure while experiencing the short-term activity, we utilized *liking* to influence their emotions based on previous research [10, 20]. Hence, the research question we are exploring:

RQ1: Will participants' pleasure be higher in *liked* VE with time-dilated items.

Next, we predict that the participants will improve their selective attention by utilizing them to select items, and they will be more focused during the activity that lasts for a short period [3, 24]. Thus, we want to know that:

RQ2: Will participants be more focused in the *selected* VE with time-dilated items.

Based on the related works, the more focused the participants are, the less attention they can share for time perception [16, 21]. Thus, we also discuss that:

RQ3: Will participants feel time passes slowly in the *selected* and *liked* VE with time-dilated items.

Lastly, we assume that liking may have a relation toward immersion [6]. Hence, we discuss about:

RQ4: Will participants be more immersed in the *liked* VE with time-dilated items.

4 Design and Implementation

To create a VE with a sense of time dilation, we prepared factors that could cause time dilation by using HTC Vive Pro. Since people have different perceptions of different lengths of time, we should find out subjective time dilation factors with the same influence by a pilot study. In the pilot study, we prepared four categories: music, environmental color, environmental picture, and objects for the VE. Each category has eight options, with 32 items. Next, we chose the items that made the participants feel time passed slowly to apply in the user study. In the user study, we separated these items by participants' liking and selection into four dimensions for exploring their time perception. Besides, to apply our research, we wanted the participants to complete a short-term activity for closer daily life. Hence, we also asked the participants to read and conclude an article in each VE.

4.1 Pilot Study: Temporal cue

Based on the literature review, we prepared 32 items in each VE (four categories with eight options) to determine the influence of subjective time dilation factors. Each VE had only one item. We expected the participants to report the prospective perception of

time that they reported their time perception during the activity but not afterward [3, 34]. The participants would go through all the VE and answer the subjective time. The criteria of four categories are (1) music [5]: Background music used in the environment, primarily consisting of low-frequency music and white noise; (2) environmental color [2]: The overall color tone of the scene overlaid on the environmental picture in the form of a filter, with each environmental color being of low brightness and medium saturation in various hues; (3) environmental picture [8]: 360-degree panoramic images, all featuring spaces with sparse content and a wider field of view; (4) slowly moving objects [38]: Dynamic objects without time cues move much slower compared to those in reality.

4.1.1 Task and Procedure. We recruited 16 participants (11 females, aged 19 to 29, Mean= 22.38, SD= 2.68). During the experiment, participants would experience the 32 VE and answer their prospective subjective time perception by Personal Time Passage Questionnaire (PTPQ) [39] after each VE. In the post-questionnaire, we used PTPQ questionnaire to collect the perceived prospective time for each environment. From the study by Crystal and Jonathon [7], it was found that users have different perceptions of different duration of time. Therefore, the participants would go through two times with different time duration(30 seconds and 1 minute) in the same VE, in a total of 64 VE.

4.1.2 Results and Discussion. The answer to the PTPQ questionnaire [39] was a 5-point Likert scale, where 5 indicates "time passed slowly" and 1 indicates "time passed quickly." There was also an option for the participants to choose when they experienced mixed perceptions of time passing quickly and slowly. We condensed these responses into four categories: 0 (time passed both quickly and slowly), 1 (time passed quickly), 2 (time passed normally), and 3 (time passed slowly). To identify elements contributing to the perception of time passing slowly, we applied two constraints:(1) We aimed to minimize instances of mixed time perception, accepting no more than three occurrences of '0' across different conditions. (2) Any result with more than five instances of '1' was excluded, indicating that the element might lead to a perception of faster time passage. Following these criteria, the filtered items were used in the user study.

5 User Study: Time Dilation

Next, filtering the items that can cause a sense of time dilation. We wanted to determine whether the participants' liking and selection would occur in time dilation or affect their pleasure and focus during the user study.

5.1 Participants and Procedure

We recruited 26 participants (13 males, aged 20-35, Mean: 22.85, SD: 3.17) with VR experience. Using 21 items filtered by the pilot study, participants classified these into "like" and "no preference," selecting at least two from each. They then chose one item from each category to help them focus, resulting in four groups: like and select (L-S), no preference and select (NP-S), like and not select (L-NS), and no preference and not select (NP-NS). These items composed the virtual environments (Figure 1). Participants followed this procedure to create four VEs. If a category had more than two

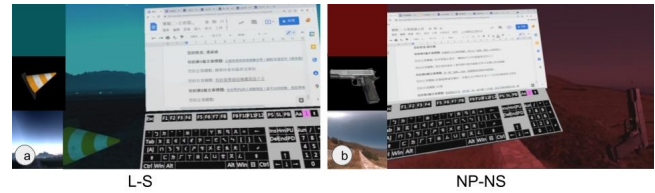


Figure 1: Example of the VE; (a) is composed by a traffic cones in mountain path with cyan-blue filter; (b) is composed by a shooting gun in a desert with red filter.

items, one was randomly chosen. The sequence of VEs was counterbalanced using a Latin square design. Participants completed a short-term activity to maintain focus, selecting three interesting and three boring articles from a list of 10 titles. They read the remaining four articles across the four VEs, ensuring no strong emotional response influenced the results. After reading each article, participants wrote their thoughts to ensure they were focused. Each environment was timed, and participants filled out a post-test questionnaire estimating the duration felt in the VE, the PTPQ [39], PAD emotional model [18], and IPQ scale [30]. We also interviewed the participants how they felt. The entire experiment lasted 30-40 minutes.

5.2 Results and Discussion

To analyze the result, we used two-way ANOVA to analyze the significant difference between the groups with repeated measurements. The variables are liking or not and selecting or not. Figure 2 shows the result, we divide our measurements into two sections, mental measurement and time perception. We observe the participant's pleasure, focus, and IPQ in the mental measurement. Next, in time perception, we discuss the participants' feelings about time by measuring their estimated time, time perception ratio, and PTPQ.

5.2.1 Mental Measurements. For the RQ1, participants with L-S (mean= 3.59, SD= 0.58) had the highest pleasure, followed by L-NS and NP-S, the lowest is NP-NS (mean= 2.37, SD= 0.70). The result shows that there is a significant effect on liking ($F= 29.98, p< 0.001$) and selection ($F= 13.78, p= 0.001$). However, there is no interaction between liking and selection. Next, for the RQ2, participants with L-S (mean= 3.58, SD= 0.86) have the highest focus, and NP-NS (mean= 2.69, SD= 1.01) has lowest Focus. The result indicates an effect on liking ($F= 6.93, p= 0.014$) and selection ($F= 4.50, p= 0.044$). However, there is no interaction between liking and selection. Last, for RQ4, participants have the highest presence in L-S (mean= 3.42, SD= 0.70), with L-NS, NP-S, and NP-NS sequentially. There is an effect on liking ($F= 4.95, p=0.035$).

5.2.2 Time Perception. Compared to the real-time, we find no statistical difference between estimated time and real-time in each VE (RQ3). We then calculated the time perception ratio ($= (\text{estimated time} - \text{real-time}) / \text{real-time}$). When the ratio is greater than zero, the user feels a longer time than in real-time and feels that time passes slowly. Participants with L-NS (mean= 0.23; SD= 0.52) have the highest ratio, with NP-S and NP-NS being next. Last for the

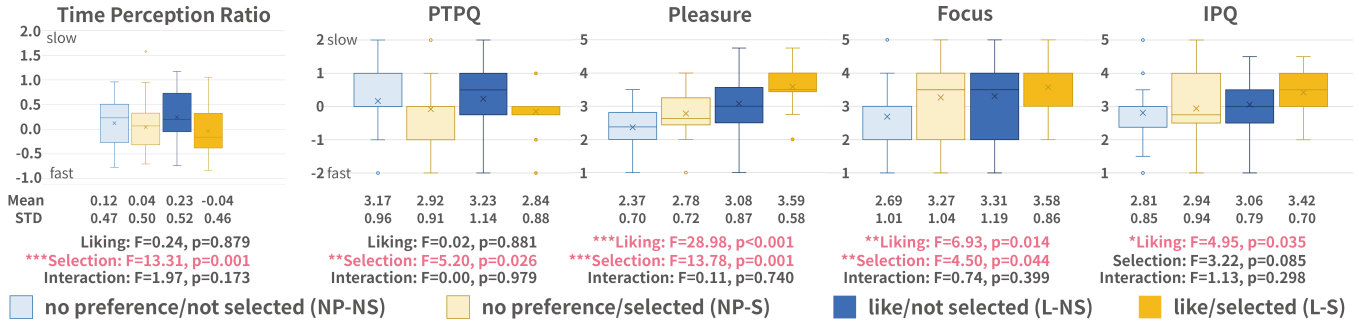


Figure 2: Result of User Study

L-S (mean= -0.04, SD= 0.46). There is a significant effect on selection ($F= 13.31, p=0.001$). PTPQ is a questionnaire that measures the participant's feelings about time, with one standing for time passing quickly and five standing for time passing slowly. The result shows that participants in the L-NS (mean= 3.23, SD= 1.14) and NP-NS (mean= 3.17, SD= 0.96) scenes felt time passed slowly, while participants in NP-S (mean= 2.92, SD= 0.91) and L-S (mean= 2.84, SD=0.88) scene felt time passed quickly. There is also a significant effect on selection ($F=5.20, p=0.026$).

5.2.3 Discussion. The result shows that liking is related to pleasure and IPQ, while selection is more related to focus. Nonetheless, despite creating a VE with time-dilated items, participants did not experience the anticipated significant time dilation. Regarding PTPQ, selection is key to the participants' subjective time perception. Our results indicate that while 'liking' and 'selection' impact participants' pleasure and focus in a time-dilated VE, they do not significantly alter their estimated time. The reason why participants felt time passed slower (PTPQ is over 3) in non-selected VEs still needs to be explored. We consider that the inefficacy might be due to the insufficient intensity of the time-dilated items filtered in our pilot study, suggesting a more rigorous design in the future.

6 Limitation and Challenge

6.1 Estimated Time in Time Dilation VE

As mentioned in the discussion, we revealed that VE with non-selected items might lead to time dilation. However, we can not state that VE with selected items causes time compression. The VE we created was intended to slow the perception of time. In this situation, while the participants felt time passed more quickly ($PTPQ < 3$, in Figure 2), this does not indicate time compression; rather, it suggests a relative quickening against a slower baseline. To know if L-S VE made participants feel time passed more quickly ($PTPQ > 3$), further research is needed using VE with items designed to make time feel normal.

6.2 Culture Impact

Time perception varies among individuals and across cultures. For example, participants in an environment colored by red light felt a sense of time compression [11]. However, our pilot study found that participants had different feelings toward red light; they felt time passed slowly. We consider that different cultures or backgrounds

might influence different factors in participants' time perception. Hence, we recommend conducting a pilot study to determine which items can influence participants' time perception due to their culture. Nonetheless, due to the discussion of our research, we suggest that the pilot study should be more rigorously designed.

6.3 Influence of Factors Integration

Our user study on VE comprised four categories, but we did not investigate how each category individually influenced the VE. Interviews revealed that 11 participants noted the impact of music on their experience, while eight mentioned color, and eight referenced the influence of reading articles. However, the specific effects of these categories were not further analyzed. Understanding how each category complements the others to form a cohesive VE requires further study. Additionally, creating a VE that adapts to different situations necessitates focusing on interactions for better design guidelines. For instance, understanding how integrated factors interact with each other and how users engage with the VE and activities is crucial. While our research proposed design considerations for reading activities in a time-dilated VE, these guidelines may not apply universally. Further research should explore these interactions to develop adaptable and effective design strategies for specific applications.

7 Summary and Future Work

The old saying, "Time flies when you're having fun." [1] Hence, we want to explore design considerations for creating an environment that can make sense of time dilation and still have positive emotions in short-term activities. In our pilot study, we filtered items that made participants feel time passed slowly. We then conducted a user study to determine whether liking and selection influence participants' time perception and mental status. The finding revealed that 'liking' and 'selection' positively impacted participants' pleasure and focus in the VE. However, they did not significantly feel a sense of time dilation in the VE. This outcome suggests the need for a more rigorous and nuanced approach in the design process toward our pilot study to manipulate time perception in VEs precisely.

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